

COS20019 - Cloud Computing Architecture

Assignment 1 - Part B

Creating and deploying Photo Album
website onto a simple AWS infrastructure

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Due Date: 15/10/2023

Assignment 1b rubric		
Criteria	Page number	Pts
Infrastructure Requirements		
VPC with 2 public and 2 private subnets	2-3	0.5 pts
Correct Public and Private Routing tables with correct subnet associations	4	1 pts
Security groups properly configured and attached.	5-7	1 pts
Network ACL properly configured and attached	8-10	1.5 pts
Correct Web server and Test instances running in correct subnets	11-15	0.5 pts
Database schema as specified	16-20	0.5 pts
Database running in correct subnets	21-22	1 pts
S3 objects publicly accessible, using proper access policy	22-23	0.5 pts
Functional Requirements		
album.php page displayed from EC2 Web server	24-25	1 pts
Provided URL is persistent (Elastic IP Association)	25-26	0.5 pts
Photos loaded from S3 with matching metadata from RDS	27-28	1 pts
Web server instance reachable from Test instance via ICMP	29	1 pts
Deductions		
Documentation not as specified or poorly presented (up to minus 20)	None	0 pts
Serious misconfigurations of AWS services being used (up to minus 20)	None	0 pts
Total points: 10		

Infrastructure Requirements

VPC with 2 public and 2 private subnets

1. Create a new VPC

- Name: TLuongVPC
- Region: us-east-1
- IPv4 CIDR block: 10.0.0.0/16

The screenshot displays the AWS Management Console interface for creating a new VPC. The breadcrumb navigation shows 'VPC > Your VPCs > Create VPC'. The main heading is 'Create VPC' with an 'Info' link. A descriptive text states: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.'

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate
TLuongVPC

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IPs
CIDR block size must be between /16 and /28.

Preview

VPC [Show details](#)
Your AWS virtual network.

TLuongVPC-vpc

Subnets (4)
Subnets within this VPC

us-east-1a
TLuongVPC-subnet-public1-us-east-1a
TLuongVPC-subnet-private1-us-east-1a

us-east-1b
TLuongVPC-subnet-public2-us-east-1b
TLuongVPC-subnet-private2-us-east-1b

2. Number of AZs

- First availability zone: us-east-1a
- Second availability zone: us-east-1b

3. Number of subnets

- Public: 2
- Private: 2
- Customise subnets CIDR blocks: Follow the diagram in the assignment specification.

The screenshot shows the AWS VPC console configuration page for a new VPC. The left pane contains configuration options, and the right pane shows a preview of the VPC and its subnets.

Configuration Options (Left Pane):

- Tenancy:** Default
- Number of Availability Zones (AZs):** 2 (Selected). Info: Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.
- Customize AZs:**
 - First availability zone:** us-east-1a
 - Second availability zone:** us-east-1b
- Number of public subnets:** 2. Info: The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.
- Number of private subnets:** 4. Info: The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.
- Customize subnets CIDR blocks:**
 - Public subnet CIDR block in us-east-1a:** 10.0.1.0/24 (256 IPs)
 - Public subnet CIDR block in us-east-1b:** 10.0.2.0/24 (256 IPs)
 - Private subnet CIDR block in us-east-1a:** 10.0.3.0/24 (256 IPs)
 - Private subnet CIDR block in us-east-1b:** 10.0.4.0/24 (256 IPs)

Preview (Right Pane):

- VPC:** TLuongVPC-vpc
- Subnets (4):**
 - us-east-1a:**
 - TLuongVPC-subnet-public1-us-east-1a
 - TLuongVPC-subnet-private1-us-east-1a
 - us-east-1b:**
 - TLuongVPC-subnet-public2-us-east-1b
 - TLuongVPC-subnet-private2-us-east-1b

Correct Public and Private Routing tables with correct subnet associations

1. Correct subnet associations

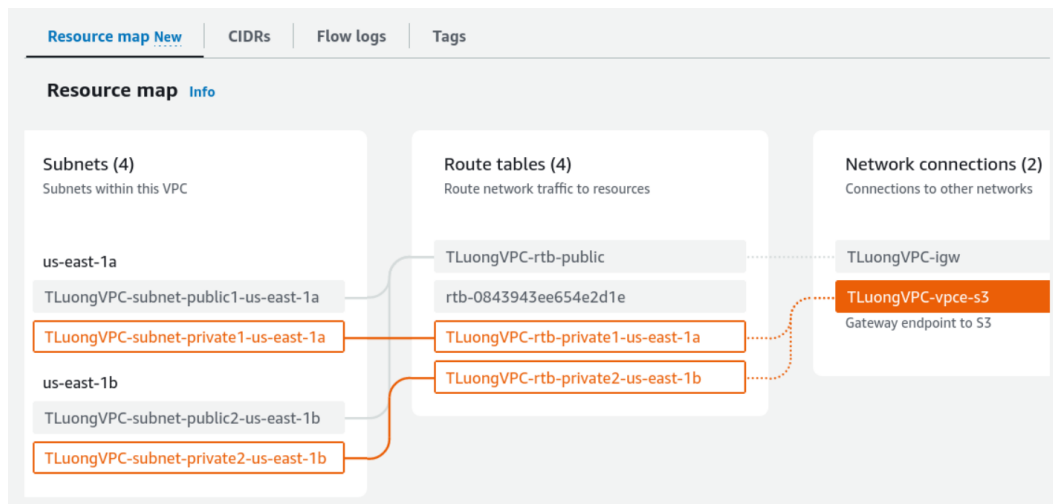
The screenshot shows the AWS VPC console configuration for a VPC named **TLuongVPC-vpc**. The configuration is as follows:

- Tenancy:** Default
- Number of Availability Zones (AZs):** 2 (us-east-1a and us-east-1b)
- Number of public subnets:** 2 (10.0.1.0/24 and 10.0.2.0/24)
- Number of private subnets:** 2 (10.0.3.0/24 and 10.0.4.0/24)

The Preview section shows the VPC and its associated subnets:

- VPC:** TLuongVPC-vpc
- Subnets (4):**
 - us-east-1a
 - TLuongVPC-subnet-public1-us-east-1a
 - TLuongVPC-subnet-private1-us-east-1a
 - us-east-1b
 - TLuongVPC-subnet-public2-us-east-1b
 - TLuongVPC-subnet-private2-us-east-1b

2. Correct routing tables



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Security groups properly configured and attached.

1. Security groups configured

a. TestInstanceSG: All traffic - Anywhere

The screenshot shows the AWS Management Console 'Create security group' page. The 'Basic details' section is filled out with the following information:

- Security group name: TestInstanceSG
- Description: Test Instance Security Group
- VPC: vpc-0e8b6fad3f2af16fb

The 'Inbound rules' section shows a single rule configured:

Type	Protocol	Port range	Source	Description - optional
All traffic	All	All	Anywhere (0.0.0.0/0)	

An 'Add rule' button is visible at the bottom left of the inbound rules section.

b. WebServerSG:

- HTTP (80) - Anywhere
- SSH (22) - Anywhere
- All ICMP - IPv4 - TestInstanceSG

The screenshot shows the AWS Management Console 'Create security group' page for WebServerSG. The 'Basic details' section is filled out with the following information:

- Security group name: WebServerSG
- Description: Web Server Security Group
- VPC: vpc-0e8b6fad3f2af16fb

The 'Inbound rules' section shows three rules configured:

Type	Protocol	Port range	Source	Description - optional
HTTP	TCP	80	Anywhere (0.0.0.0/0)	
SSH	TCP	22	Anywhere (0.0.0.0/0)	
All ICMP - IPv4	ICMP	All	TestInstanceSG (sg-0912ed108d1d3e955)	

An 'Add rule' button is visible at the bottom left of the inbound rules section.

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c. DBServerSG

i. MySQL (3306) - WebServerSG

The screenshot shows the AWS Management Console for a Security Group named 'DBServerSG'. The 'Inbound rules' section is active, showing a rule for MySQL/Aurora on port 3306, allowing traffic from the source 'sg-0587f4018c8cab5d1'. The rule is currently being edited.

Security group name: DBServerSG
Description: Database Server Security Group
VPC: vpc-0e8b6fad3f2af16fb

Inbound rules:

Type	Protocol	Port range	Source	Description - optional
MySQL/Aurora	TCP	3306	Cu...	

Add rule

2. Security groups attached

a. WebServerSG attached to Bastion Instance EC2

The screenshot shows the AWS Management Console for EC2 instances. The 'Instances (1/2)' page is displayed, showing two instances: 'Bastion Instance' and 'Test Instance'. The 'Bastion Instance' is selected, and its details are shown, including the security group 'sg-0587f4018c8cab5d1 (WebServerSG)'.

Instances (1/2)

Name	Instance ID	Instance state	Instance type	Status check	Alarm
Bastion Instance	i-011a39b09b3181be6	Running	t2.micro	Initializing	No alarm
Test Instance	i-039d05821facfb079	Running	t2.micro	Initializing	No alarm

Instance: i-011a39b09b3181be6 (Bastion Instance)

Security details

IAM Role	Owner ID	Launch time
-	841427755345	Wed Oct 04 2023 12:42:51 GMT+0700 (Indochina Time)

Security groups

Security group
sg-0587f4018c8cab5d1 (WebServerSG)

Inbound rules

Name	Security group rule ID	Port range	Protocol	Source
-	sgr-06a79f2486f5fd1eb	All	ICMP	sg-0912ed108d1d3e955
-	sgr-04165c7f61d2a6de4	22	TCP	0.0.0.0/0
-	sgr-03d5270c09bc9c47c	80	TCP	0.0.0.0/0

b. TestInstanceSG attached to Test Instance EC2

The screenshot shows the AWS Management Console interface. On the left is a navigation menu with categories like EC2 Dashboard, Instances, Images, Elastic Block Store, and Network & Security. The main content area is titled 'Instances (1/2) Info'. It features a search bar and a filter 'Instance state = running'. A table lists two instances: 'Bastion Instance' and 'Test Instance'. The 'Test Instance' is selected, and its details are shown below. The details include tabs for Details, Security, Networking, Storage, Status checks, Monitoring, and Tags. The 'Security' tab is active, showing 'Security details' with IAM Role as '-', Owner ID as '841427755345', and Security groups as 'sg-0912ed108d1d3e955 (TestInstanceSG)'. It also shows 'Inbound rules' with a table containing one rule.

Name	Instance ID	Instance state	Instance type	Status check	Alarm
Bastion Instance	i-011a39b09b3181be6	Running	t2.micro	Initializing	No alarm
Test Instance	i-039d05821facfb079	Running	t2.micro	Initializing	No alarm

Name	Security group rule ID	Port range	Protocol	Source
-	sgr-0709f547864c1909f	All	All	0.0.0.0/0

c. DBServerSG attached to assignment1b-testdb RDS

The screenshot shows the AWS Management Console interface for the Amazon RDS service. The left navigation menu includes options like Dashboard, Databases, Query Editor, Performance Insights, and Snapshots. The main content area is titled 'assignment1b-testdb'. It features a 'Summary' section with a table showing DB identifier, CPU usage, Status (Available), Class, Role, Current activity, Engine, and Region & AZ. Below the summary are tabs for Connectivity & security, Monitoring, Logs & events, Configuration, Maintenance & backups, and Tags. The 'Connectivity & security' tab is active, showing a table with Endpoint & port, Networking, and Security details.

DB identifier	CPU	Status	Class
assignment1b-testdb	2.77%	Available	db.t3.micro

Endpoint & port	Networking	Security
Endpoint: assignment1b-testdb.ckti4ybwtgj.us-east-1.rds.amazonaws.com	Availability Zone: us-east-1a VPC: TLuongVPC-vpc	VPC security groups: DBServerSG (sg-084b7c5f511f17c36) - Active

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Network ACL properly configured and attached

1. Create new network ACL: PublicSubnet2NACL

VPC > Network ACLs > Create network ACL

Create network ACL [Info](#)

A network ACL is an optional layer of security that acts as a firewall for controlling traffic in and out of a subnet.

Network ACL settings

Name - optional
Creates a tag with a key of 'Name' and a value that you specify.

PublicSubnet2NACL

VPC
VPC to use for this network ACL.

vpc-0e8b6fad3f2af16fb (TLuongVPC-vpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q PublicSubnet2NACL X Remove tag

Add tag

You can add 49 more tags

Cancel Create network ACL

2. Edit inbound rules

aws Services Search [Alt+S] N. Virginia voclabs/user2752985=103488117@student.swin.edu.au @ 8414-2775...

VPC > Network ACLs > acl-055a2a31f6a23af5d / PublicSubnet2NACL > Edit inbound rules

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the VPC.

Rule number Info	Type Info	Protocol Info	Port range Info	Source Info	Allow/Deny Info	
1	SSH (22)	TCP (6)	22	0.0.0.0/0	Allow	Remove
2	All ICMP - IPv4	ICMP (1)	All	10.0.4.0/24	Allow	Remove
3	HTTP (80)	TCP (6)	80	0.0.0.0/0	Allow	Remove
4	HTTPS (443)	TCP (6)	443	0.0.0.0/0	Allow	Remove
5	All TCP	TCP (6)	All	10.0.3.0/24	Allow	Remove
6	All TCP	TCP (6)	All	10.0.4.0/24	Allow	Remove
*	All traffic	All	All	0.0.0.0/0	Deny	

Add new rule Sort by rule number

Cancel Preview changes Save changes

3. Edit outbound rules

The screenshot shows the 'Edit outbound rules' page in the AWS Management Console. The breadcrumb trail is 'VPC > Network ACLs > acl-055a2a31f6a23af5d / PublicSubnet2NACL > Edit outbound rules'. The page title is 'Edit outbound rules' with an 'Info' link. A note states: 'Outbound rules control the outgoing traffic that's allowed to leave the VPC.' Below this is a table of rules. The first rule has a rule number of 1, type 'All traffic', protocol 'All', port range 'All', destination '0.0.0.0/0', and action 'Allow'. The second rule has a rule number of 2, type 'All traffic', protocol 'All', port range 'All', destination '0.0.0.0/0', and action 'Deny'. At the bottom, there are buttons for 'Add new rule', 'Sort by rule number', 'Cancel', 'Preview changes', and 'Save changes'.

Rule number	Type	Protocol	Port range	Destination	Allow/Deny	
1	All traffic	All	All	0.0.0.0/0	Allow	Remove
2	All traffic	All	All	0.0.0.0/0	Deny	

4. Edit subnet association

Choose Public Subnet 2

- AZ: us-east-1b
- IPv4CIDR: 10.0.2.0/24

The screenshot shows the 'Edit subnet associations' page in the AWS Management Console. The breadcrumb trail is 'VPC > Network ACLs > acl-055a2a31f6a23af5d / PublicSubnet2NACL > Edit subnet associations'. The page title is 'Edit subnet associations' with an 'Info' link. A note states: 'Change which subnets are associated with this network ACL.' Below this is a table of available subnets. The first three subnets are unselected, and the fourth is selected. The selected subnet is 'TLuongVPC-subnet-public2-us-east-1b' with Subnet ID 'subnet-0a19adcd401a3b5b8', Associated with 'acl-025ff55a346c82d20', Availability Zone 'us-east-1b', and IPv4 CIDR '10.0.2.0/24'. At the bottom, there are buttons for 'Cancel' and 'Save changes'.

Name	Subnet ID	Associated with	Availability Zone	IPv4 CIDR
☐ TLuongVPC-subnet-private1-us-east-1a	subnet-0a0c22d997b83...	acl-025ff55a346c82d20	us-east-1a	10.0.3.0/24
☐ TLuongVPC-subnet-public1-us-east-1a	subnet-067904b5153cb...	acl-025ff55a346c82d20	us-east-1a	10.0.1.0/24
☐ TLuongVPC-subnet-private2-us-east-1b	subnet-0f547d983630a...	acl-025ff55a346c82d20	us-east-1b	10.0.4.0/24
☒ TLuongVPC-subnet-public2-us-east-1b	subnet-0a19adcd401a3...	acl-025ff55a346c82d20	us-east-1b	10.0.2.0/24

Confirm association

The screenshot shows the AWS Management Console interface for Network ACLs. The left sidebar contains navigation links for VPC dashboard, EC2 Global View, and various VPC resources. The main content area displays a table of Network ACLs. The 'PublicSubnet2NACL' is selected, and its details are shown below the table.

Name	Network ACL ID	Associated with	Default	VPC ID
-	acl-025ff55a346c82d20	3 Subnets	Yes	vpc-0e8b6f...
☒ PublicSubnet2NACL	acl-055a2a31f6a23af5d	subnet-0a19adcd401a3b5b8 / TLuongVPC-sub...	No	vpc-0e8b6f...
-	acl-09fa489987a9873be	6 Subnets	Yes	vpc-035d8...

acl-055a2a31f6a23af5d / PublicSubnet2NACL

Details

Network ACL ID	Associated with	Default	VPC ID
acl-055a2a31f6a23af5d	subnet-0a19adcd401a3b5b8 / TLuongVPC-subnet-public2-us-east-1b	No	vpc-0e8b6fad3f2af16fb / TLuongVPC-vpc
Owner	841427755345		

Correct Web server and Test instances running in correct subnets

1. Create a web server instance

- Name: Bastion Instance
- AMI: Amazon Linux 2 AMI (HVM), SSD Volume Type
- Instance type: t2.micro

Launch an instance | EC2

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Services Search [Alt+S] N. Virginia voclabs/user2752985=103488117@student.swin.edu.au @ 8414-2775...

Name
Bastion Instance [Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

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Quick Start

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Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type [Free tier eligible](#)
ami-0bb4c991fa89d4b9b (64-bit (x86)) / ami-0a445ece583184891 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

▼ Instance type [Info](#)

Instance type

t2.micro [Free tier eligible](#)
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.0716 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour
[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations [Compare instance types](#)

▼ Summary

Number of instances [Info](#)
1

Software Image (AMI)
[Amazon Linux 2023.2.2...read more](#)
ami-067d1e60475437da2

Virtual server type (instance type)
t2.micro

Firewall (security group)
[New security group](#)

Storage (volumes)
1 volume(s) - 8 GiB

Cancel [Launch instance](#) [Review commands](#)

Name: Trac Duc Anh Luong - ID: 103488117

- VPC: TLuongVPC
- Subnet: TLuongVPC-subnet-public2-us-east-1b
- Security group: WebServerSG

aws Services Search [Alt+S] N. Virginia voclabs/user2752985=103488117@student.swin.edu.au @ 8414-2775...

Key pair name - required
LTDA [Create new key pair](#)

Network settings [Info](#)

VPC - required [Info](#)
vpc-0e8b6fad3f2af16fb (TLuongVPC-vpc) [Create new VPC](#)

Subnet [Info](#)
subnet-0a19adcd401a3b5b8 TLuongVPC-subnet-public2-us-east-1b [Create new subnet](#)
VPC: vpc-0e8b6fad3f2af16fb Owner: 841427755345
Availability Zone: us-east-1b IP addresses available: 251 CIDR: 10.0.2.0/24

Auto-assign public IP [Info](#)
Disable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
Select security groups
WebServerSG sg-0587f4018c8cab5d1 [Compare security group rules](#)
VPC: vpc-0e8b6fad3f2af16fb

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2 Kernel 5.10 AMI...[read more](#)
ami-0bb4c991fa89d4b9b

Virtual server type (instance type)
t2.micro

Firewall (security group)
WebServerSG

Storage (volumes)
1 volume(s) - 8 GiB

Cancel **Launch instance** [Review commands](#)

- Advanced details: paste the bash script from assignment 1a to the user data text area

The screenshot shows the AWS Management Console 'Create Instance' page. The 'Advanced details' section is expanded, showing various configuration options. The 'User data' field contains a bash script for installing and configuring a web server. The 'Summary' section on the right shows the instance configuration: 1 instance, Amazon Linux 2 Kernel 5.10 AMI, t2.micro instance type, WebServerSG security group, and 1 volume of 8 GiB. The 'Launch instance' button is highlighted in orange.

Kernel ID [Info](#)
Select

Nitro Enclave [Info](#)
Select
Nitro Enclaves are not compatible with instance types that have less than 2 vCPUs.

License configurations [Info](#)
Select

☐ Specify CPU options
The selected instance type does not support CPU options.

Metadata accessible [Info](#)
Select

Metadata transport
Select

Metadata version [Info](#)
Select

Metadata response hop limit [Info](#)
Select

Allow tags in metadata [Info](#)
Select

User data - optional [Info](#)
Upload a file with your user data or enter it in the field.
[Choose file](#)

```
#!/bin/bash
yum update -y
amazon-linux-extras install -y lamp-mariadb10.2-php7.2 php7.2
service httpd start
yum install -y httpd mariadb-server php-mbstring php-xml
systemctl start httpd
systemctl enable httpd
usermod -a -G apache ec2-user
chown -R ec2-user:apache /var/www
chmod 2775 /var/www
find /var/www -type d -exec sudo chmod 2775 {} \;
find /var/www -type f -exec sudo chmod 0664 {} \;
echo "<?php echo '<h2>Welcome to COS80001. Installed PHP version: ',
phpversion() . '</h2>'; ?>" > /var/www/html/phpinfo.php
```

☐ User data has already been base64 encoded

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2 Kernel 5.10 AMI...[read more](#)
ami-0bb4c991fa89d4b9b

Virtual server type (instance type)
t2.micro

Firewall (security group)
WebServerSG

Storage (volumes)
1 volume(s) - 8 GiB

Cancel **Launch instance**
[Review commands](#)

Name: Trac Duc Anh Luong - ID: 103488117

2. Create a test instance

- Name: Bastion Instance
- AMI: Amazon Linux 2 AMI (HVM), SSD Volume Type
- Instance type: t2.micro

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
Test Instance [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

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Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type [Free tier eligible](#)
ami-0bb4c991fa89d4b9b (64-bit (x86)) / ami-0a445ece583184891 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Description
Amazon Linux 2 Kernel 5.10 AMI 2.0.20230926.0 x86_64 HVM gp2

Architecture **AMI ID**
64-bit (x86) ami-0bb4c991fa89d4b9b [Verified provider](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2 Kernel 5.10 AMI...[read more](#)
ami-0bb4c991fa89d4b9b

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Review commands](#)

Instance type

t2.micro [Free tier eligible](#)
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.0716 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour
[Additional costs apply for AMIs with pre-installed software](#)

[All generations](#) [Compare instance types](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
LTDA [Create new key pair](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2 Kernel 5.10 AMI...[read more](#)
ami-0bb4c991fa89d4b9b

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Name: Trac Duc Anh Luong - ID: 103488117

- VPC: TLuongVPC
- Subnet: TLuongVPC-subnet-private2-us-east-1b
- Security group: TestInstanceSG

aws Services Search [Alt+S] N. Virginia voclabs/user2752985=103488117@student.swin.edu.au @ 8414-2775...

Key pair name - required
LTDA [Create new key pair](#)

Network settings [Info](#)

VPC - required [Info](#)
vpc-0e8b6fad3f2af16fb (TLuongVPC-vpc) 10.0.0.0/16 [Create new VPC](#)

Subnet [Info](#)
subnet-0f547d983630a9671 TLuongVPC-subnet-private2-us-east-1b
VPC: vpc-0e8b6fad3f2af16fb Owner: 841427755345
Availability Zone: us-east-1b IP addresses available: 251 CIDR: 10.0.4.0/24 [Create new subnet](#)

Auto-assign public IP [Info](#)
Disable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
Select security groups
TestInstanceSG sg-0912ed108d1d3e955 [Compare security group rules](#)
VPC: vpc-0e8b6fad3f2af16fb

Security groups that you add or remove here will be added to or removed from all your network interfaces.

[Advanced network configuration](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2 Kernel 5.10 AMI...[read more](#)
ami-0bb4c991fa89d4b9b

Virtual server type (instance type)
t2.micro

Firewall (security group)
TestInstanceSG

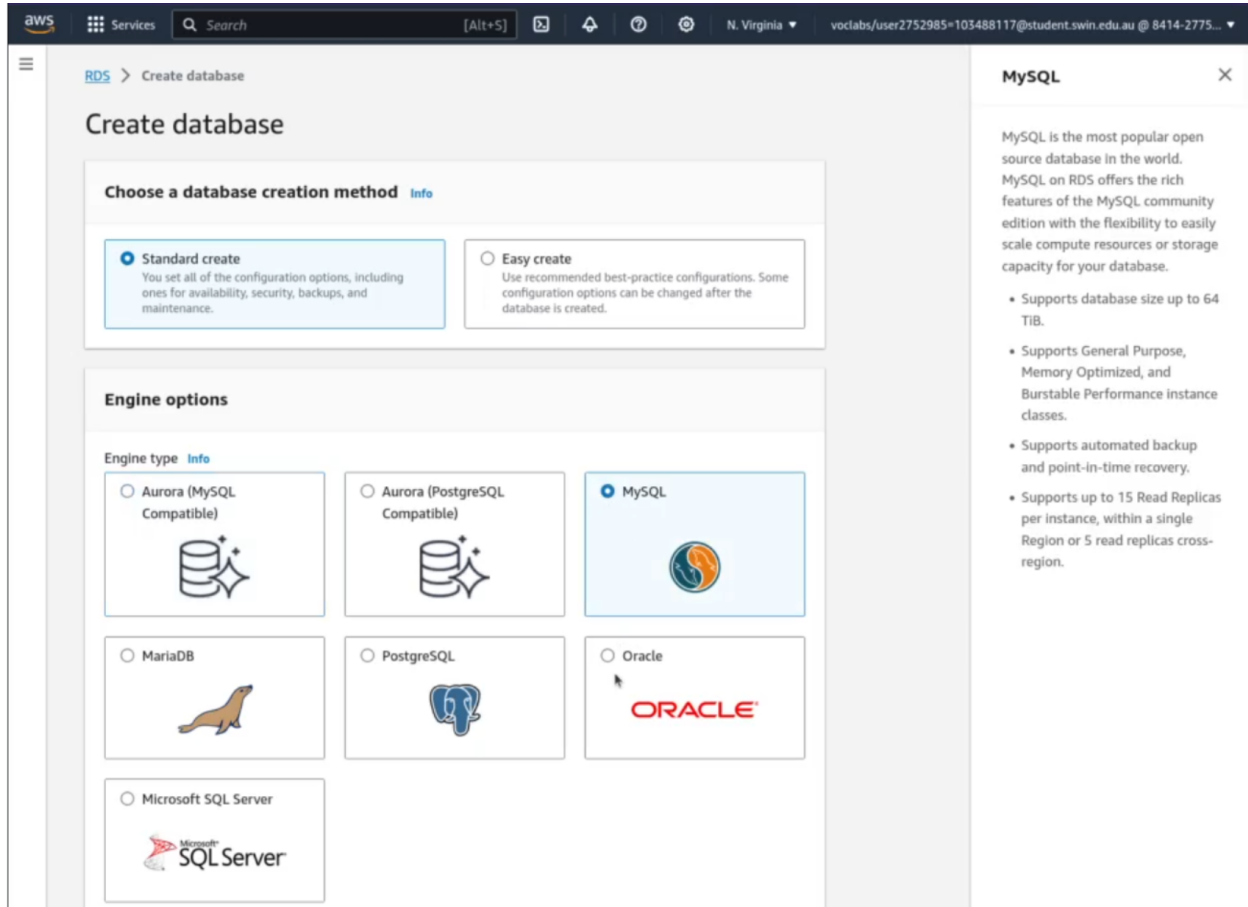
Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Review commands](#)

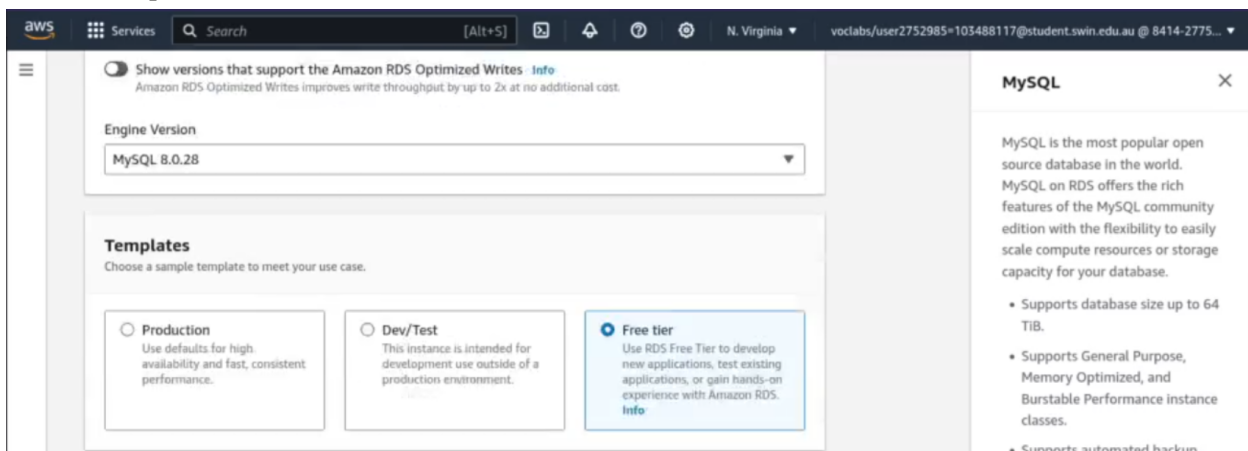
Database schema as specified

1. Create database

- Creation method: standard create
- Engine type: MySQL



- Engine version: 8.0.28 (there was no 8.0.25 option like in the assignment specification)
- Templates: Free tier



- DB instance identifier: assignment1b-testdb

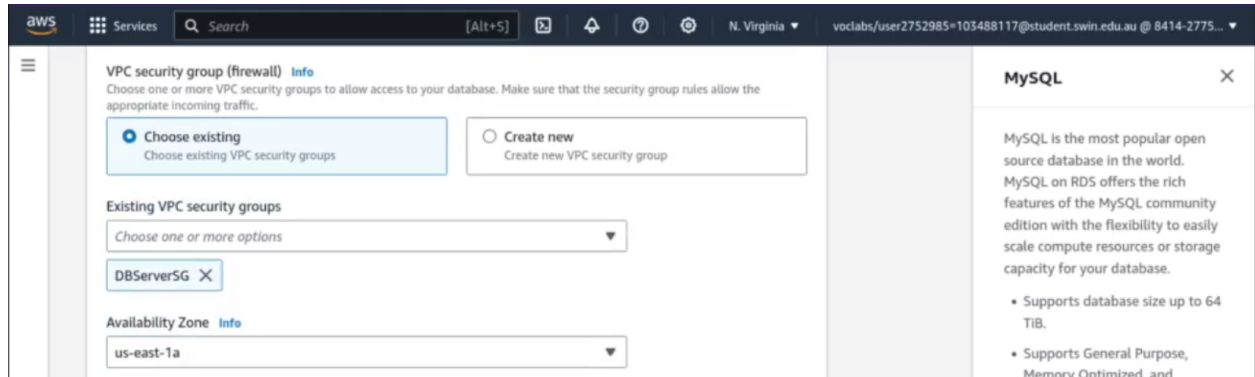
The screenshot shows the AWS Management Console for a new MySQL DB instance. The 'Settings' tab is active. Under 'DB instance identifier', the name 'assignment1b-testdb' is entered. A note states: 'The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.'

- Compute resource: Don't connect to an EC2 compute resource (Do this later)
- Network type: IPv4
- VPC: TLuongVPC
- DB subnet group: dbsubnetgroup (create this before creating the database, I will provide the details in the next requirement)
- Public access: No

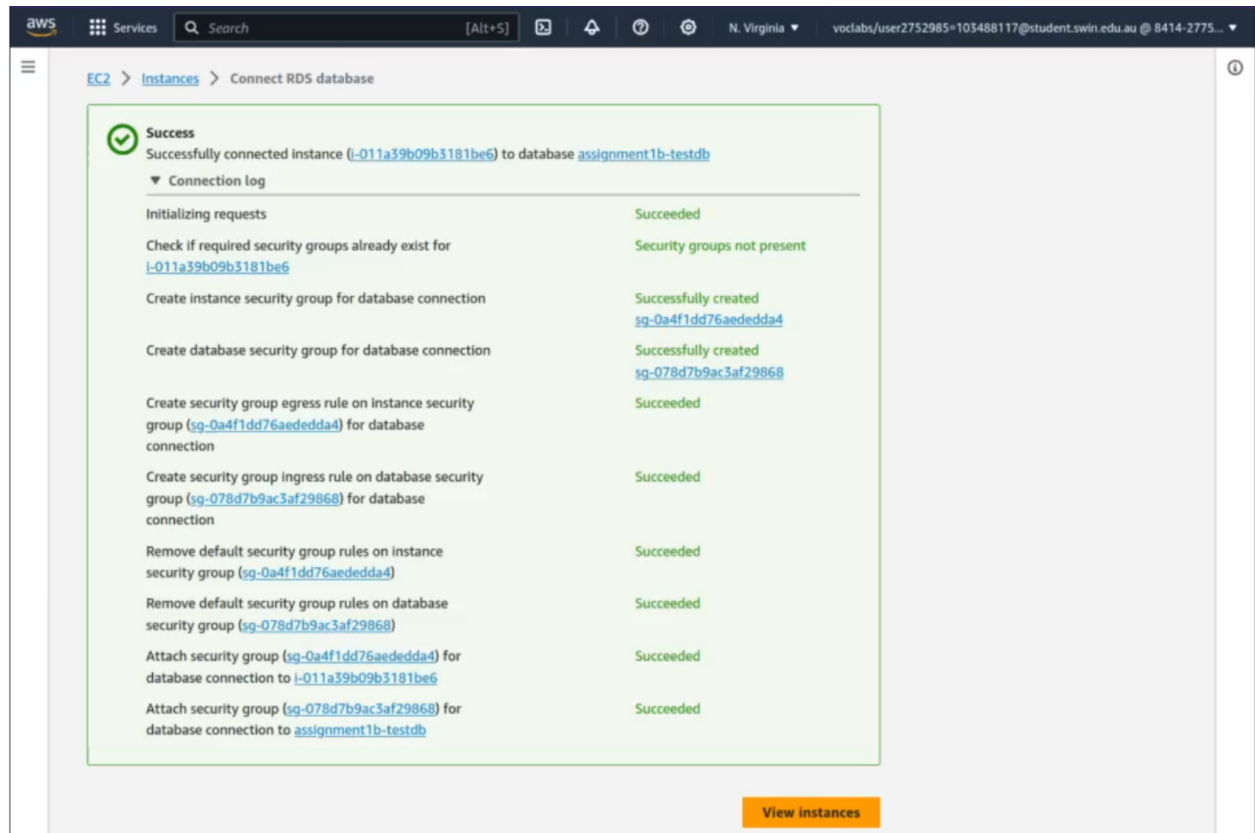
The screenshot shows the 'Compute resource' and 'Network type' settings for a new MySQL DB instance. Under 'Compute resource', 'Don't connect to an EC2 compute resource' is selected. Under 'Network type', 'IPv4' is selected. The 'Virtual private cloud (VPC)' is set to 'TLuongVPC-vpc (vpc-0e8b6fad3f2af16fb)'. The 'DB subnet group' is set to 'dbsubnetgroup'. Under 'Public access', 'No' is selected.

Name: Trac Duc Anh Luong - ID: 103488117

- VPC security group: DBServerSG
- AZ: us-east-1a



2. Connect EC2 Bastion Instance to the created RDS database



3. Install PhpMyAdmin in the EC2 Bastion Instance

```
ec2-user@ip-10-0-2-166:/var/www/html
inflatng: phpMyAdmin-4.8.2-english/vendor/twig/twig/src/TokenParser/UseTokenParser.php
inflatng: phpMyAdmin-4.8.2-english/vendor/twig/twig/src/TokenParser/WlthTokenParser.php
inflatng: phpMyAdmin-4.8.2-english/vendor/twig/twig/src/TokenStream.php
inflatng: phpMyAdmin-4.8.2-english/vendor/twig/twig/src/TwigFilter.php
inflatng: phpMyAdmin-4.8.2-english/vendor/twig/twig/src/TwigFunction.php
inflatng: phpMyAdmin-4.8.2-english/vendor/twig/twig/src/TwigTest.php
creating: phpMyAdmin-4.8.2-english/vendor/twig/twig/src/Util/
inflatng: phpMyAdmin-4.8.2-english/vendor/twig/twig/src/Util/DeprecationCollector.php
inflatng: phpMyAdmin-4.8.2-english/vendor/twig/twig/src/Util/TemplateDirIterator.php
inflatng: phpMyAdmin-4.8.2-english/version_check.php
inflatng: phpMyAdmin-4.8.2-english/view_create.php
inflatng: phpMyAdmin-4.8.2-english/view_operations.php
inflatng: phpMyAdmin-4.8.2-english/yarn.lock
finishing deferred symbolic links:
phpMyAdmin-4.8.2-english/vendor/bin/highlight-query -> ../phpmyadmin/sql-parser/bin/highlight-query
phpMyAdmin-4.8.2-english/vendor/bin/lint-query -> ../phpmyadmin/sql-parser/bin/lint-query
[ec2-user@ip-10-0-2-166 html]$ mv phpMyAdmin-4.8.2-english phpmyadmin
[ec2-user@ip-10-0-2-166 html]$
```

4. Edit the config.sample.inc.php file and change the host from localhost to the RDS endpoint

```
30 /* Server parameters */
31 $cfg['Servers'][$i]['host'] = 'assignment1b-testdb.ckti4ybwtgj.us-east-1.rds.amazonaws.com';
32 $cfg['Servers'][$i]['compress'] = false;
33 $cfg['Servers'][$i]['AllowNoPassword'] = false;
```

5. Rename the config file to config.inc.php

Remote site: /var/www/html/phpmyadmin

- spool
- tmp
- www
 - cgi-bin
 - html
 - cos80001
 - phpmyadmin
 - yp

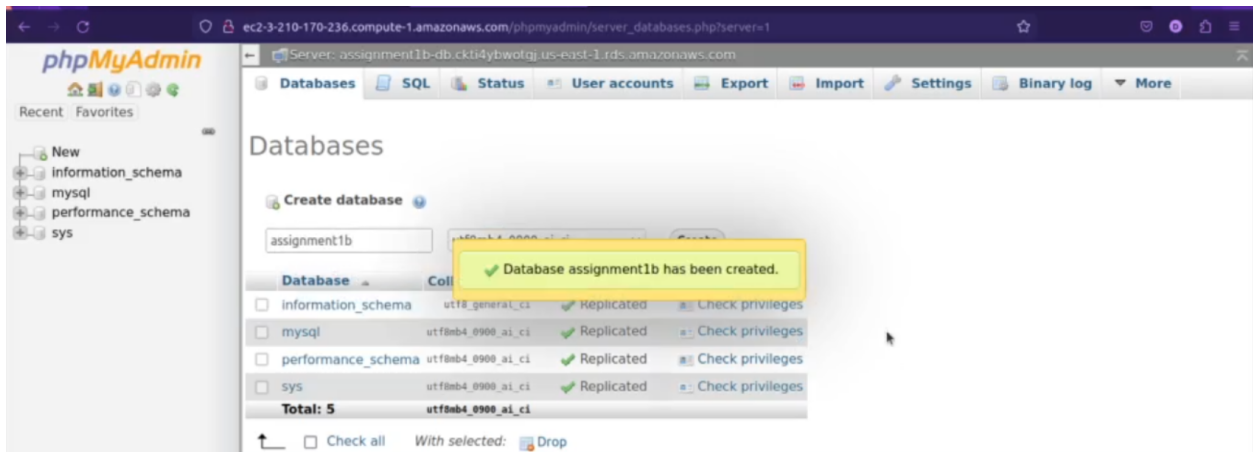
Filename	Filesize	Filetype	Last modified	Permission	Owner/Grou
composer.json	3.2 KB	json-file	06/21/2018 ...	-rw-r--r--	ec2-user ...
composer.lock	93.6 KB	lock-file	06/21/2018 ...	-rw-r--r--	ec2-user ...
config.inc.php	4.6 KB	php-file	10/04/2023 ...	-rw-r--r--	ec2-user ...

Name: Trac Duc Anh Luong - ID: 103488117

6. Login to PhpMyAdmin using the public dns + /phpmyadmin/ link, the credential has already been defined in the RDS creation step.

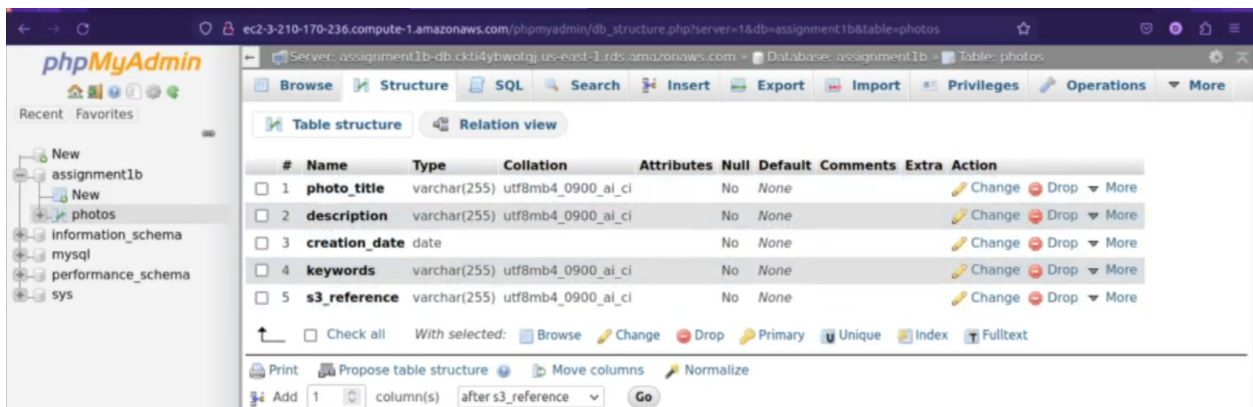


7. Create database assignment1b



8. Create table photos with the required columns:

- photo_title: varchar(255)
- description: varchar(255)
- creation_date: date
- keywords: varchar(255)
- s3_reference: varchar(255)



Database running in correct subnets

1. Create RDS Subnet group

- Name: DBSubnetGroup
- VPC: TLuongVPC
- AZs: us-east-1a, us-east-1b
- Subnets: private subnets (10.0.3.0/24 and 10.0.4.0/24)

Amazon RDS

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.
DBSubnetGroup
Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description
Private Subnet Group

VPC
Choose a VPC Identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.
TLuongVPC-vpc (vpc-0e8b6fad3f2af16fb)

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.
Choose an availability zone
us-east-1a X us-east-1b X

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.
Select subnets
subnet-0a0c22d997b83a670 (10.0.3.0/24) X
subnet-0f547d983630a9671 (10.0.4.0/24) X

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet ID	CIDR block
us-east-1a	subnet-0a0c22d997b83a670	10.0.3.0/24
us-east-1b	subnet-0f547d983630a9671	10.0.4.0/24

Cancel Create

2. Connect the RDS database to the subnet group (Done in the RDS creation step)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Network type [Info](#)

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

TLuongVPC-vpc (vpc-0e8b6fad3f2af16fb)
4 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

dbsubnetgroup
2 Subnets, 2 Availability Zones

Public access [Info](#)

☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

MySQL

MySQL is the most popular open source database in the world. MySQL on RDS offers the rich features of the MySQL community edition with the flexibility to easily scale compute resources or storage capacity for your database.

- Supports database size up to 64 TiB.
- Supports General Purpose, Memory Optimized, and Burstable Performance instance classes.
- Supports automated backup and point-in-time recovery.
- Supports up to 15 Read Replicas per instance, within a single Region or 5 read replicas cross-region.

S3 objects publicly accessible, using proper access policy

1. Create bucket

- Name: 103488117-assignment1b
- Region: us-east-1

Create bucket [Info](#)

Buckets are containers for data stored in S3. [Learn more](#)

General configuration

Bucket name

103488117-assignment1b

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

AWS Region

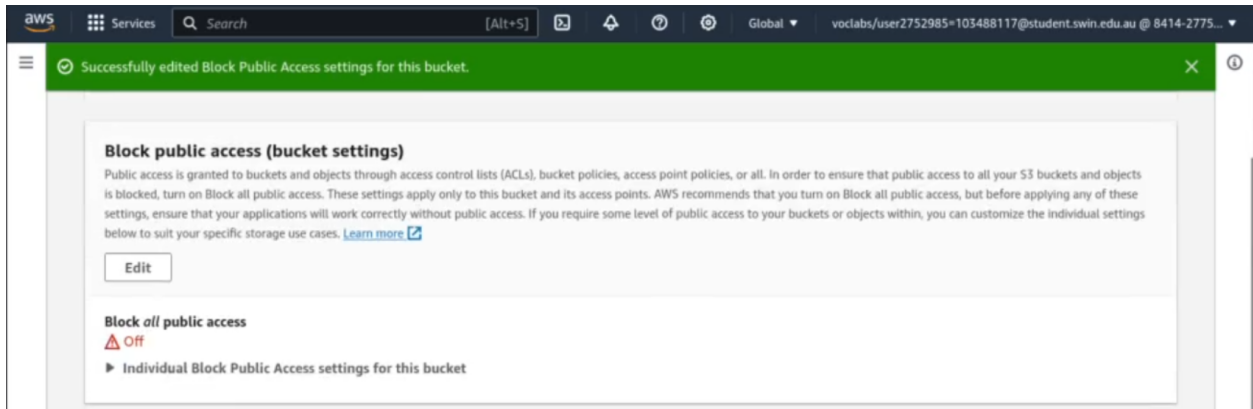
US East (N. Virginia) us-east-1

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

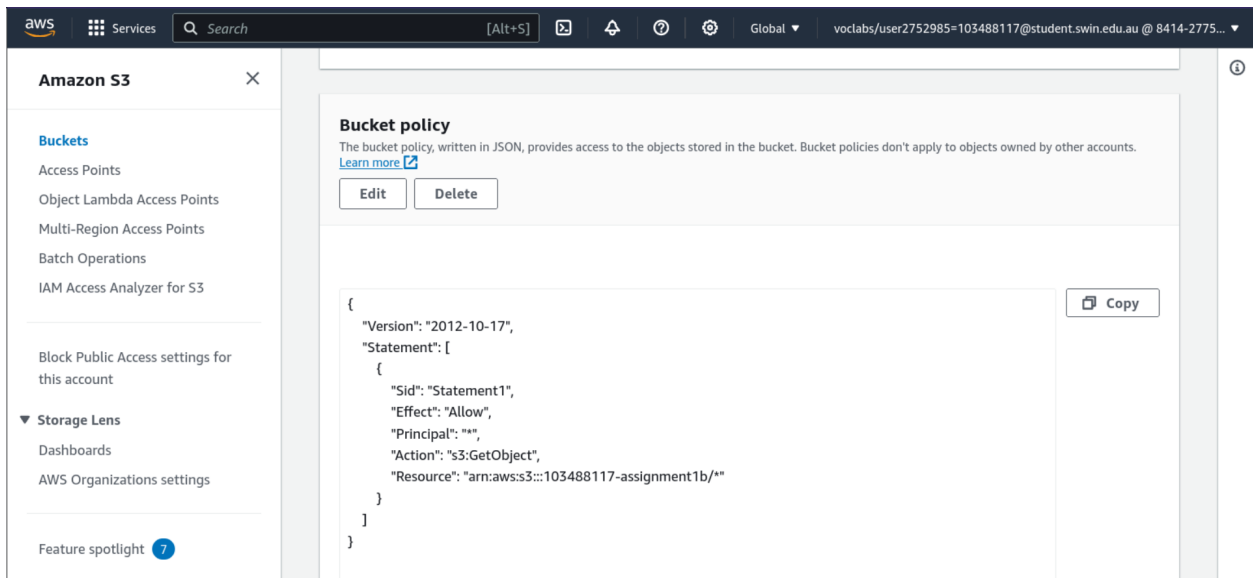
[Choose bucket](#)

2. Turn off block all public access



3. Edit bucket policy

- Effect: Allow
- Principal: *
- Action: s3:GetObject
- Resource: [Bucket_ARN]/* (all resources)



Functional Requirements

album.php page displayed from EC2 Web server

1. Change the constants.php file to your information

```
constants.php X
constants.php > ...
33 */
34
35 // [ACTION REQUIRED] your full name
36 define('STUDENT_NAME', 'Trac Duc Anh Luong');
37 // [ACTION REQUIRED] your Student ID
38 define('STUDENT_ID', '103488117');
39 // [ACTION REQUIRED] your tutorial session
40 define('TUTORIAL_SESSION', 'Saturday 10:00 AM');
41
42 // [ACTION REQUIRED] name of the S3 bucket that stores images
43 define('BUCKET_NAME', '103488117-assignment1b');
44 // [ACTION REQUIRED] region of the above bucket
45 define('REGION', 'us-east-1');
46 // no need to update this const
47 define('S3_BASE_URL', 'https://'.BUCKET_NAME.'.s3.amazonaws.com/');
48
49 // [ACTION REQUIRED] name of the database that stores photo meta-data (note that this is not the DB identifier of the RDS in
50 define('DB_NAME', 'assignment1b');
51 // [ACTION REQUIRED] endpoint of RDS instance
52 define('DB_ENDPOINT', 'assignment1b-testdb.ckt4dybwotgj.us-east-1.rds.amazonaws.com');
53 // [ACTION REQUIRED] username of your RDS instance
54 define('DB_USERNAME', 'admin');
55 // [ACTION REQUIRED] password of your RDS instance
56 define('DB_PWD', 'ducanh2003');
57
58 // [ACTION REQUIRED] name of the DB table that stores photo's meta-data
59 define('DB_PHOTO_TABLE_NAME', 'photos');
60 // The table above has 5 columns:
61 // [ACTION REQUIRED] name of the column in the above table that stores photo's titles
62 define('DB_PHOTO_TITLE_COL_NAME', 'photo_title');
63 // [ACTION REQUIRED] name of the column in the above table that stores photo's descriptions
64 define('DB_PHOTO_DESCRIPTION_COL_NAME', 'description');
65 // [ACTION REQUIRED] name of the column in the above table that stores photo's creation dates
66 define('DB_PHOTO_CREATIONDATE_COL_NAME', 'creation_date');
67 // [ACTION REQUIRED] name of the column in the above table that stores photo's keywords
68 define('DB_PHOTO_KEYWORDS_COL_NAME', 'keywords');
69 // [ACTION REQUIRED] name of the column in the above table that stores photo's links in S3
70 define('DB_PHOTO_S3REFERENCE_COL_NAME', 's3_reference');
71 ?>
```

2. Transfer the files to the Ec2 Bastion Instance

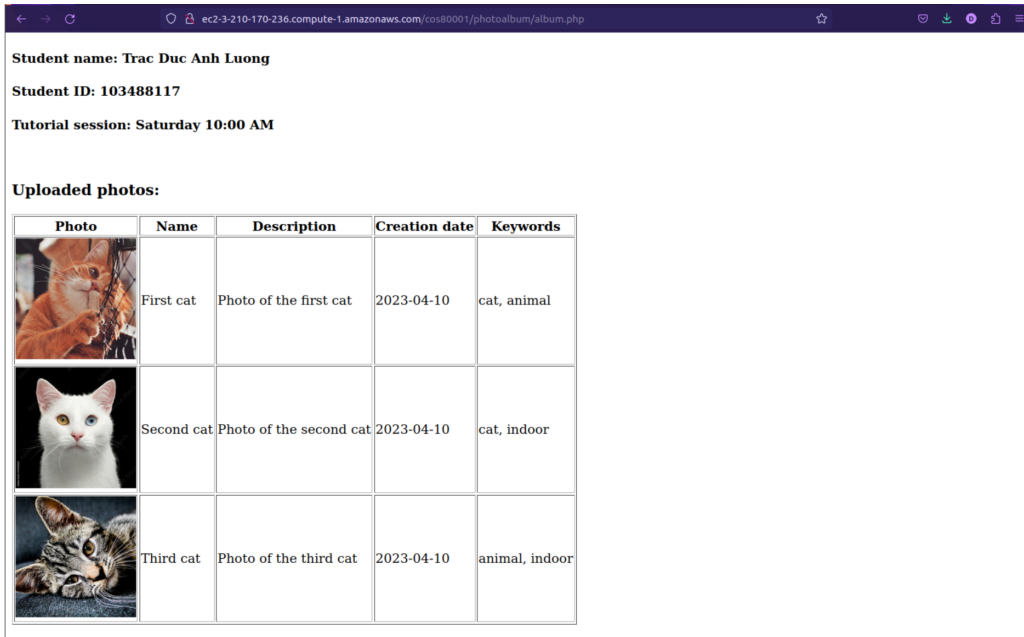
Remote site: /var/www/html/cos80001/photoalbum

- tmp
- www
 - cgi-bin
 - html
 - cos80001
 - photoalbum
 - phpmyadmin
 - yp

Filename ^	Filesize	Filetype	Last modified	Permission	Owner/Grou
..					
album.php	1.1 KB	php-file	10/04/2023 ...	-rw-rw-r--	ec2-user ...
constants.php	3.0 KB	php-file	10/04/2023 ...	-rw-rw-r--	ec2-user ...
defaultstyle.css	388 B	css-file	10/04/2023 ...	-rw-rw-r--	ec2-user ...
mydb.php	1.3 KB	php-file	10/04/2023 ...	-rw-rw-r--	ec2-user ...
photo.php	1.3 KB	php-file	10/04/2023 ...	-rw-rw-r--	ec2-user ...

Name: Trac Duc Anh Luong - ID: 103488117

3. album.php page displayed from EC2 Bastion Instance



Student name: Trac Duc Anh Luong
Student ID: 103488117
Tutorial session: Saturday 10:00 AM

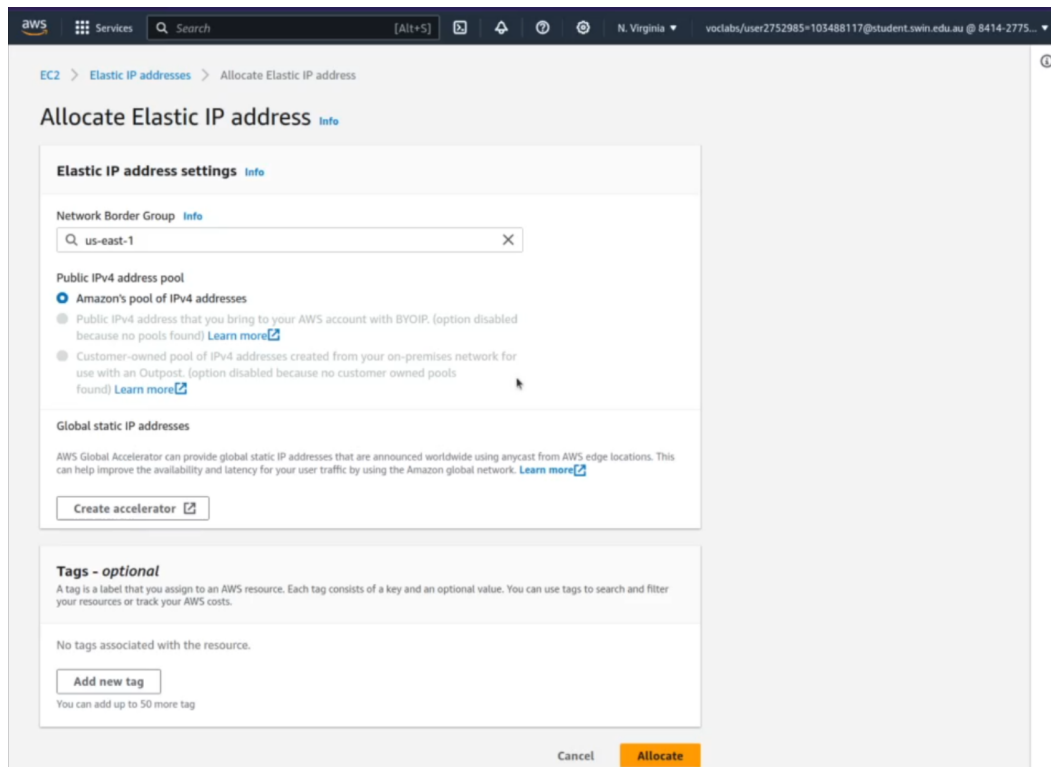
Uploaded photos:

Photo	Name	Description	Creation date	Keywords
	First cat	Photo of the first cat	2023-04-10	cat, animal
	Second cat	Photo of the second cat	2023-04-10	cat, indoor
	Third cat	Photo of the third cat	2023-04-10	animal, indoor

Provided URL is persistent (Elastic IP Association)

1. Allocate Elastic Ip address

- Network Border Group: us-east-1



EC2 > Elastic IP addresses > Allocate Elastic IP address

Allocate Elastic IP address [Info](#)

Elastic IP address settings [Info](#)

Network Border Group [Info](#)
us-east-1

Public IPv4 address pool

- ☒ Amazon's pool of IPv4 addresses
- ☐ Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) [Learn more](#)
- ☐ Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) [Learn more](#)

Global static IP addresses

AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)

[Create accelerator](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tag

[Cancel](#) [Allocate](#)

Name: Trac Duc Anh Luong - ID: 103488117

2. Associate Elastic IP address

- Instance: Bastion Instance

Elastic IP address: 3.210.170.236

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

Warning: If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance

Choose an instance

- i-011a39b09b3181be6 (Bastion Instance) - running
- i-039d05821facb079 (Test Instance) - running

Choose a private IP address

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.

☐ Allow this Elastic IP address to be reassociated

Cancel Associate

3. Bastion Instance now has public IPv4 address and public IPv4 DNS

Name	Instance ID	Instance state	Instance type	Status check	Alarm
Bastion Instance	i-011a39b09b3181be6	Running	t2.micro	Initializing	No ala
Test Instance	i-039d05821facb079	Running	t2.micro	Initializing	No ala

Instance: i-011a39b09b3181be6 (Bastion Instance)

Networking details

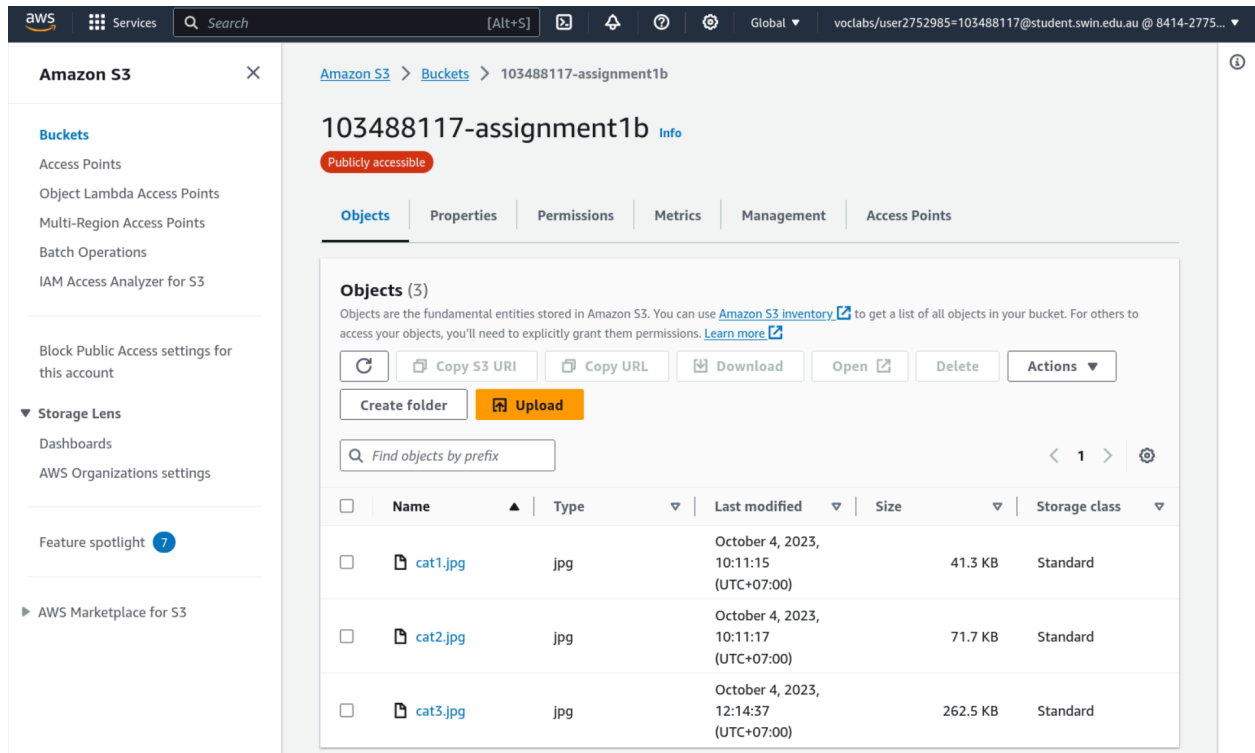
Public IPv4 address 3.210.170.236 open address	Private IPv4 addresses 10.0.2.166	VPC ID vpc-0e8b6fad3f2af16fb (TLuongVPC-vpc)
Public IPv4 DNS ec2-3-210-170-236.compute-1.amazonaws.com open address	Private IP DNS name (IPv4 only) ip-10-0-2-166.ec2.internal	

4. Public IPv4 address: 3.210.170.236

5. Website URL: ec2-3-210-170-236.compute-1.amazonaws.com/cos80001/photoalbum/album.php

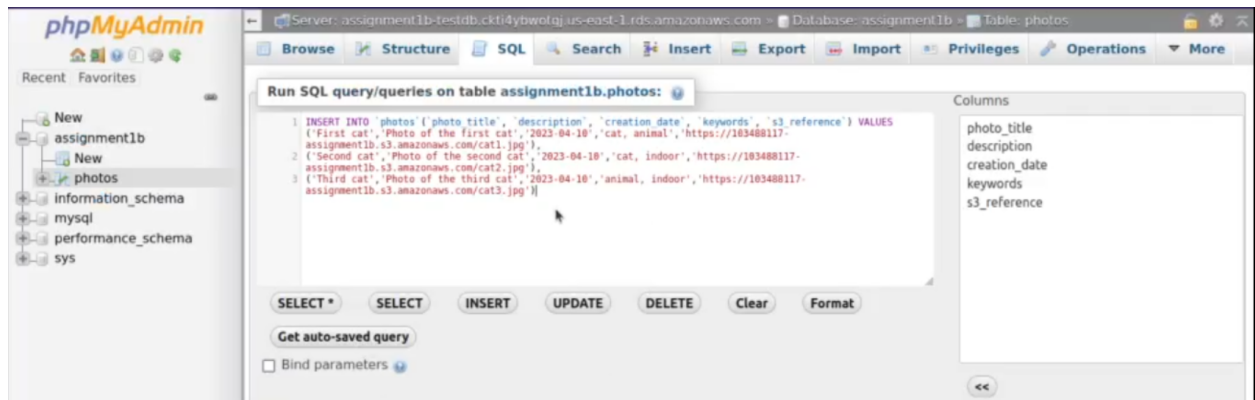
Photos loaded from S3 with matching metadata from RDS

1. Upload photos to S3



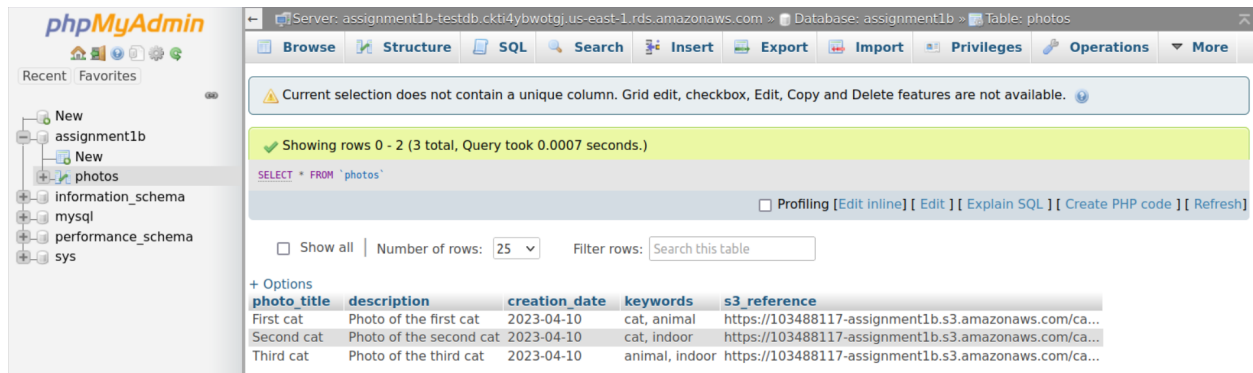
2. Insert SQL statements to the database table

s3_reference will be the link to the images in the S3 bucket.



Name: Trac Duc Anh Luong - ID: 103488117

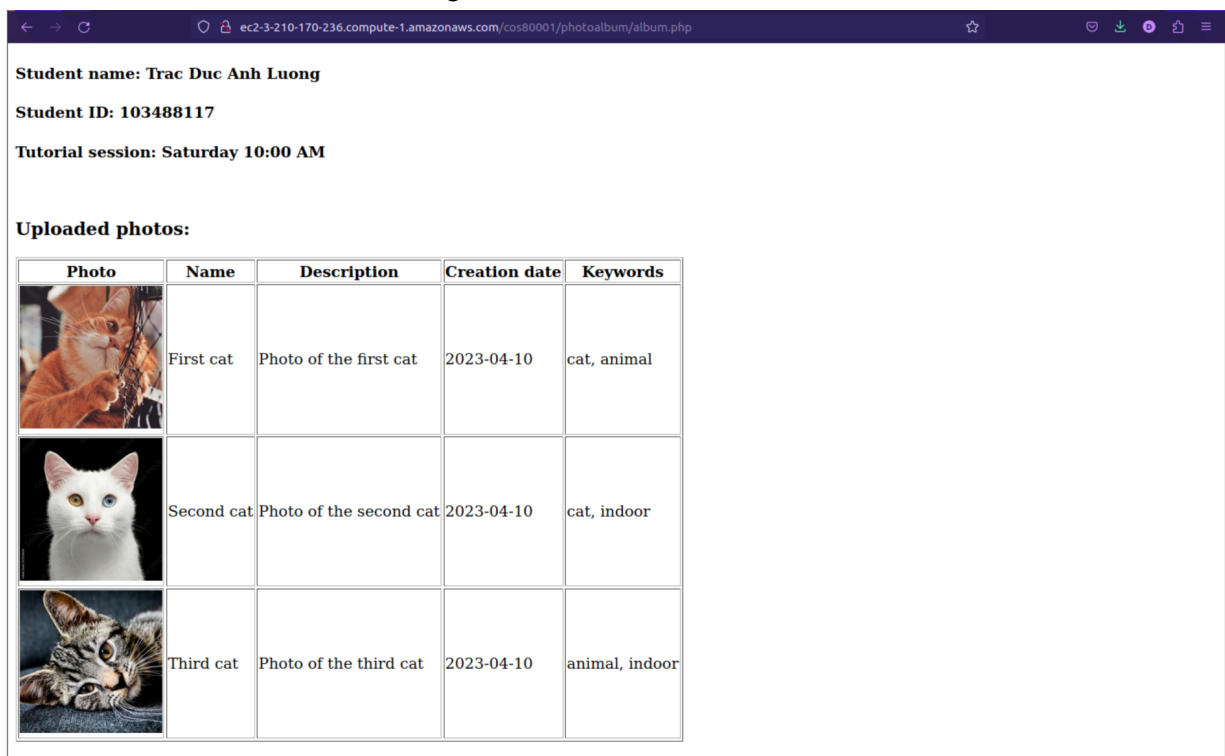
3. Database records



The screenshot shows the phpMyAdmin interface for the 'assignment1b' database. The 'photos' table is selected, and the SQL query 'SELECT * FROM `photos`' is displayed. The table contains three rows of data, each representing a cat photo with its title, description, creation date, keywords, and S3 reference.

photo_title	description	creation_date	keywords	s3_reference
First cat	Photo of the first cat	2023-04-10	cat, animal	https://103488117-assignment1b.s3.amazonaws.com/ca...
Second cat	Photo of the second cat	2023-04-10	cat, indoor	https://103488117-assignment1b.s3.amazonaws.com/ca...
Third cat	Photo of the third cat	2023-04-10	animal, indoor	https://103488117-assignment1b.s3.amazonaws.com/ca...

4. Photos loaded from S3 with matching metadata from RDS



The screenshot shows a web application interface. At the top, it displays the student's name, ID, and tutorial session. Below this, there is a section titled 'Uploaded photos:' which contains a table with three rows of photo data. Each row includes a photo thumbnail, the photo's name, description, creation date, and keywords.

Student name: Trac Duc Anh Luong
Student ID: 103488117
Tutorial session: Saturday 10:00 AM

Uploaded photos:

Photo	Name	Description	Creation date	Keywords
	First cat	Photo of the first cat	2023-04-10	cat, animal
	Second cat	Photo of the second cat	2023-04-10	cat, indoor
	Third cat	Photo of the third cat	2023-04-10	animal, indoor

Web server instance reachable from Test instance via ICMP

1. Ssh to Bastion Instance (10.0.2.166)

```
~/Desktop/COS20019 ~$ ssh -i LDA-linux.pem ec2-user@ec2-3-210-170-236.compute-1.amazonaws.com  
Last login: Wed Oct 4 04:48:47 2023 from 116.96.47.180  
  
#  
~\##### Amazon Linux 2  
~~~\#####  
~~~~\###| AL2 End of Life is 2025-06-30.  
~~~~\#/V~!_>  
~~~~A newer version of Amazon Linux is available!  
~~~~_.  
~~~~/_m/'_//Amazon Linux 2023, GA and supported until 2028-03-15.  
~~~~_/m/'_//https://aws.amazon.com/linux/amazon-linux-2023/
```

2. Ssh to Test Instance (10.0.2.210) from Bastion Instance

[illegible]

3. From Test Instance, ping Bastion Instance and provide the result

```
[ec2-user@ip-10-0-4-210 ~]$ ping 10.0.2.166
PING 10.0.2.166 (10.0.2.166) 56(84) bytes of data:
64 bytes from 10.0.2.166: icmp_seq=1 ttl=255 time=0.434 ms
64 bytes from 10.0.2.166: icmp_seq=2 ttl=255 time=0.567 ms
64 bytes from 10.0.2.166: icmp_seq=3 ttl=255 time=0.546 ms
64 bytes from 10.0.2.166: icmp_seq=4 ttl=255 time=0.551 ms
64 bytes from 10.0.2.166: icmp_seq=5 ttl=255 time=0.657 ms
64 bytes from 10.0.2.166: icmp_seq=6 ttl=255 time=0.583 ms
64 bytes from 10.0.2.166: icmp_seq=7 ttl=255 time=0.603 ms
64 bytes from 10.0.2.166: icmp_seq=8 ttl=255 time=0.532 ms
64 bytes from 10.0.2.166: icmp_seq=9 ttl=255 time=0.542 ms
64 bytes from 10.0.2.166: icmp_seq=10 ttl=255 time=0.553 ms
^C
--- 10.0.2.166 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9210ms
rtt min/avg/max/mdev = 0.434/0.556/0.657/0.061 ms
[ec2-user@ip-10-0-4-210 ~]$
```